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Percept™ (PC and RC) Neurostimulators with BrainSense™ Technology

DBS Sensing White Paper

Percept™ (PC and RC) Neurostimulators with BrainSense™ Technology

Supplemental Information for Sensing Only Configuration

Contents

Introduction

The Percept™ family of neurostimulators (Models B35200 Percept™ PC and B35300 Percept™ RC) with BrainSense™ technology† are Medtronic's next generation of implantable devices for delivering deep brain stimulation (DBS). These devices are compatible with existing Medtronic DBS products including:

- 1x4 leads (Models 3387, 3389, and 3391‡)
- 1x4 extensions (Models 37085 and 37086)
- SenSight™ directional (1-3-3-1§) leads (Models B33005 and B33015)
- SenSight™ extension (Model B34000)
- Programming/telemetry devices and other accessories

The Percept™ neurostimulators include a feature capable of bioelectric data recording (passive brain sensing†) of local field potential (LFP) signals of interest (SOI) through one or two leads implanted in the brain.

Purpose

This document provides supplemental information about the functionality and data captured with the BrainSense™ feature while using Percept™ PC and Percept™ RC neurostimulation systems. The targeted audience includes clinicians and researchers interested in post-processing brain sensing data, from patients with any of the 5 treated indications, outside of the clinical interaction with the patient. Refer to product labeling, including the A610 Clinician Programming Guide (M013075C001), for specific information including indications, safety, and warnings.

Screen shots included in the document are from the product demo software.

Sensing Feature Detailed Descriptions

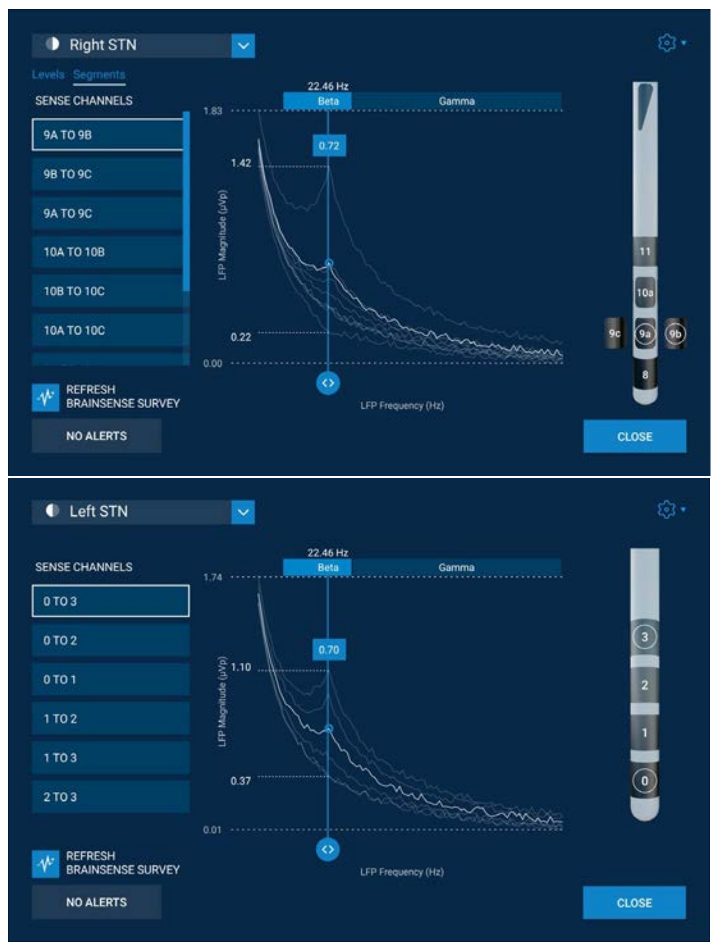
BrainSense™ Survey

Purpose: Broad spatial overview of LFP signals measurable in either hemisphere of the patient while stimulation is off.

Used: In-clinic. An electrode "Levels" measurement is approximately 40 seconds, while an electrode "Segments" measurement duration is approximately 70 seconds. Note that the "Segments" measurement can only be performed on Medtronic SenSight™ directional leads.

Detailed Design:

- User Interface displays data in the Frequency Domain: LFP Magnitude (µVp) vs. Frequency (Hz)



† The sensing feature of the Percept™ PC and Percept™ RC system is intended for use in patients receiving DBS where chronically-recorded bioelectric data may provide useful, objective information regarding patient clinical status. The majority of patients with Parkinson's disease have an identifiable signal. Signal may not be present or measurable in patients treated for essential tremor, dystonia*, epilepsy or obsessive-compulsive disorder*.

‡ Model 3391 leads are indicated for obsessive-compulsive disorder* (OCD) only and therefore not considered compatible with Model B35300 Percept RC neurostimulator, which is not indicated for OCD.

§ Electrode levels 0 and 3 are full rings, electrode levels 1 and 2 are split into 3 isolated electrodes to enable directional stimulation.

Medtronic's DBS Therapy is approved for 5 indications: Parkinson's disease, essential tremor, dystonia*, obsessive-compulsive disorder* (OCD), and epilepsy. Device indications vary, refer to product labeling.

***Humanitarian Device:** The effectiveness of the devices for the treatment of dystonia and obsessive-compulsive disorder has not been demonstrated.

- Each LFP graph represents a **differential** signal between two contacts (up to 15 pairs for each SenSight™ lead - 6 electrode ring level pairs and 9 segment-to-segment pairs, and 6 ring level pairs per 1x4 lead). Therefore, the results are measures of how similar or different two areas of the brain are:
  - Large peaks indicate electrode-pairs that are in electrically different regions of the brain.
  - Small or absent peaks indicate electrodes that are in electrically similar regions of the brain.
  - The largest peak indicates which contact pairs will provide the largest signal for sensing.
- The Session Report also contains the BrainSense™ Survey Results in the Frequency Domain.
- Stimulation is off during the measurement.
- To maximize the chance of seeing LFP signals, consider influential factors (such as medication state).
- An impedance test will be performed if not already done. Sense channels with potential shorts (<250 ohms for 1x4 leads, <350 ohms for SenSight™ Leads) or opens (>10 Kohms) are excluded.
- Screening for artifacts (cardiac, motion) is performed. Sense channels with potential artifacts are excluded. The artifact screening can be turned off to view results – see Artifact Override.
- The survey is conducted per hemisphere for 1x4 leads. For SenSight™ leads, the survey is conducted per hemisphere and per segment pair groupings or electrode level pair groupings. The time domain data (i.e. LFP Magnitude (µVp) vs. time, sampled at 250Hz) for all channels is available in the JSON file only.
  - For 1x4 leads, when the survey is conducted for the first hemisphere, e.g. Left STN, all electrode pair combinations for that lead (0-1, 0-2, 0-3, 1-2, 1-3, 2-3) are recorded simultaneously. If a second lead is present, another survey must be performed for the second hemisphere, e.g. Right STN. This second survey will simultaneously record all electrode pair combinations for the second lead (8-9, 8-10, 8-11, 9-10, 9-11, 10-11).
  - For SenSight™ leads, after a hemisphere is chosen to survey, e.g. Left STN, a grouping of electrode pairs must also be chosen, consisting of either all electrode level pairs (0-1[a,b,c], 0-2[a,b,c], 0-3, 1[a,b,c]-2[a,b,c], 1[a,b,c]-3, 2[a,b,c]-3) or all segment-to-segment pairs. If electrode level pairs are chosen, all combinations are recorded simultaneously. If segment-to-segment pairs are chosen, the survey is conducted in two passes:

    Pass 1: 1a-1b, 1b-1c, 1a-1c, 2a-2b, 2b-2c, 2a-2c

    Pass 2: 1a-2a, 1b-2b, 1c-2c

    Two surveys, an electrode level survey and a segment-to-segment survey, must be performed to record all available electrode pair combinations for a SenSight™ lead in one hemisphere. If a second SenSight™ lead is present, e.g. Right STN, both an electrode level pairs survey and a segment-to-segment pairs survey must be performed to record all available electrode pair combinations for that hemisphere. The electrode level pairs survey will record all combinations simultaneously (8-9[a,b,c], 8-10[a,b,c], 8-11, 9[a,b,c]-10[a,b,c], 9[a,b,c]-11, 10[a,b,c]-11) and the segment-to-segment survey will be conducted in two passes, similarly as described above:

    Pass 1: 9a-9b, 9b-9c, 9a-9c, 10a-10b, 10b-10c, 10a-10c

    Pass 2: 9a-10a, 9b-10b, 9c-10c
- For each channel, the time domain data (about 20 seconds of data) is converted to the frequency domain using a Fast Fourier Transform (FFT). The frequency bins are 0.98 Hz wide with centers from 0-96.68 Hz. Both the time domain and frequency domain data are available in the JSON export.
- There is an advanced “Record Streaming” option on the same screen. This allows the user to stream for longer durations, while recording LFP on stimulation-compatible electrode pairs (0-2, 0-3, 1-3, 8-10, 8-11, and 9-11 for 1x4 leads; 0-2[a,b,c], 0-3, 1[a,b,c]-3, 8-10[a,b,c], 8-11, and 9[a,b,c]-11 for SenSight™ leads). Stimulation will be off for the duration of data collection.
  - The tablet screen must be kept awake during streaming. For a Percept™ PC INS, user interaction is required to keep the screen awake (at least every 10 minutes if the screen timeout of the tablet operating system is left at the default 10-minute value). For a Percept™ RC INS, the clinician app will keep the screen awake for an additional hour while streaming, after

which user interaction at least every 10 minutes is required.

- The time domain data (i.e. LFP Magnitude (µVp) vs. time, sampled at 250Hz) is available via JSON in an Indefinite Streaming structure. Data collected via the Record Streaming (Indefinite Streaming) is not displayed on the clinician tablet or included in the Session Report.

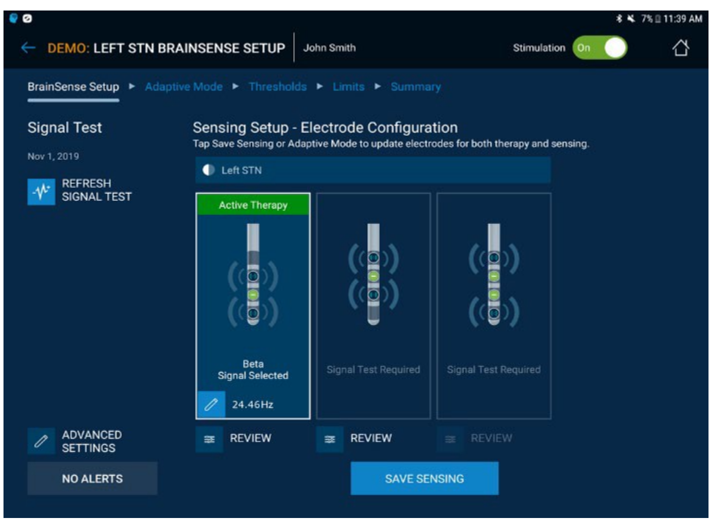
BrainSense™ Setup

Purpose: In order to use BrainSense™ technology in all manners besides BrainSense™ Survey, the user must first setup LFP sensing using BrainSense™ Setup and select a frequency band of interest (approximately 5Hz wide) to track chronically for a particular patient.

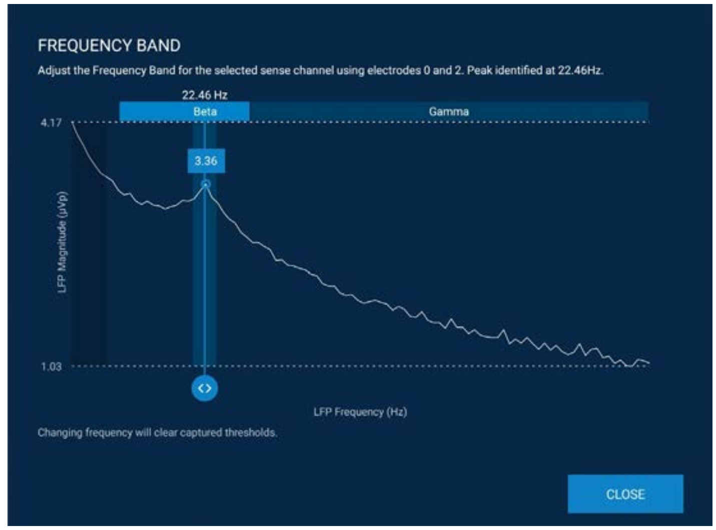
Used: In-clinic, with approximately 90 seconds measurement for setup.

Detailed Design:

- The user initiates a signal test which:
  - Performs an impedance test if not already done. Sense channels with potential shorts (<250 ohms for 1x4 leads, <350 ohms for SenSight™ Leads) or opens (>10 Kohms) are excluded.
  - Detects artifacts, both with stimulation off and on.
  - Automatically selects the largest peak on a particular channel, if that peak is in the beta or gamma frequency range and exceeds a value of 1.1 µVp, as the frequency band.
- The user then defines which electrodes to use for sensing and which to use for stimulation (3 possibilities)



- The user then optionally defines/adjusts the desired frequency band to record.



- The user interface and Session Report only indicate if artifacts were detected in the recording. The raw time domain data used for this determination is available in the JSON Export. See - Sensing Specific Data Structures.
- Optionally, the user can configure LFP Thresholds
  - Thresholds are used to categorize the LFP data relative to two clinician defined Thresholds. The LFP data can then later be viewed as above/below/between these thresholds.
  - Streaming and Timeline (see following sections) screens auto-scale their vertical axis to include threshold values, which may increase the utility of these screens.
  - To Setup Thresholds, stimulation amplitude is used to actuate the measured LFP amplitude using the following steps:
    - Set the stimulation amplitude to elicit a larger LFP signal (e.g. low stimulation amplitude where efficacy is first observed, if the signal of interest is in the beta band of a PD patient), and then capture the Upper LFP Threshold.
    - Set the stimulation amplitude to elicit a smaller LFP signal (e.g. larger stimulation amplitude with efficacy but no side effect, if the signal of interest is in the beta band of a PD patient), and then capture the Lower LFP Threshold.

    Note: Consider factors that may influence the LFP signal when setting thresholds, such as medication.

BrainSense™ Timeline

Purpose: Once BrainSense™ Setup has been completed, the Timeline is used to analyze the out-of-office data when the patient returns to the clinic. This is used to assess the data for changes in LFP activity that may occur over the course of a day(s).

Used: Outside-clinic, records LFP data chronically for follow-up data analysis in-clinic via the Timeline view.

Detailed Design:

- When the patient leaves the clinic, BrainSense™ LFP power domain data is continuously recorded while a BrainSense™ configured group is active. The neurostimulator measures raw time domain data sampled at 250Hz, converts this to the frequency domain, and then measures the power in the clinician specified band. The clinician specified band is approximately 5 Hz wide.
- If LFP thresholds were configured in this group, these also appear on the timeline view on the same plot as the recorded LFP data.
- The average stimulation amplitude, and patient limits (if configured) are also recorded and displayed to the user.
- The LFP power and the stimulation amplitudes are the average value measured over a 10 minute interval and these averages are recorded to the neurostimulator memory.

- Percept™ PC devices allow for capture of up to 60 days of LFP and stimulation data, while Percept™ RC allows for up to 35 days of data capture. If new timeline data is collected after an INS has already collected the maximal amount of data, then the oldest day will be overwritten unless BrainSense™ is turned off or the user changes to a group without BrainSense™ configured.
- Patient Marked Events, see BrainSense™ Events, also appear on the Timeline view in pink.
- Stimulation changes including Group Switch, Stim On/Off, CP Session, MRI Mode, Impedance Tests, Recharge Sessions (Percept™ RC only), and Therapy Unavailable events (Percept™ RC only) also appear on the Timeline view in green.

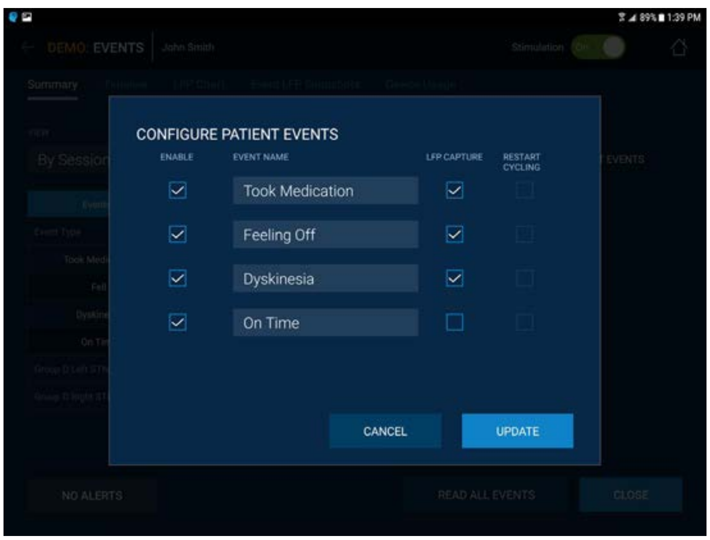
BrainSense™ Events

Purpose: Once BrainSense™ Setup has been completed, BrainSense™ Events, a.k.a LFP Snapshots can be recorded at a moment in time, showing the magnitude of the LFP signal over a range of frequencies. The LFP Snapshot is recorded when the patient records an event as configured by the clinician. Used to assess the occurrence of clinician-defined events, and associate LFP activity with those events.

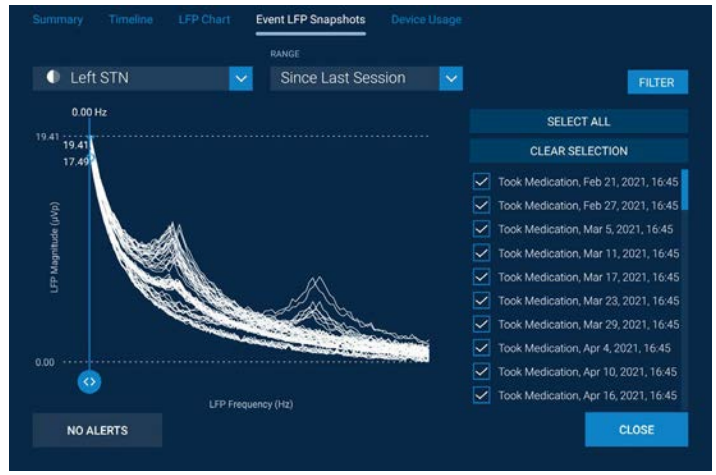
Used: Outside-clinic, with approximately 30 seconds of LFP measurement directly after patient marking an event.

Detailed Design:

- To capture LFP Snapshots when the patient records an event:
  - The active group must be one which has BrainSense™ configured.
  - The clinician must have configured Patient Events (up to 4), with "LFP Capture" selected as an option when the event is configured.



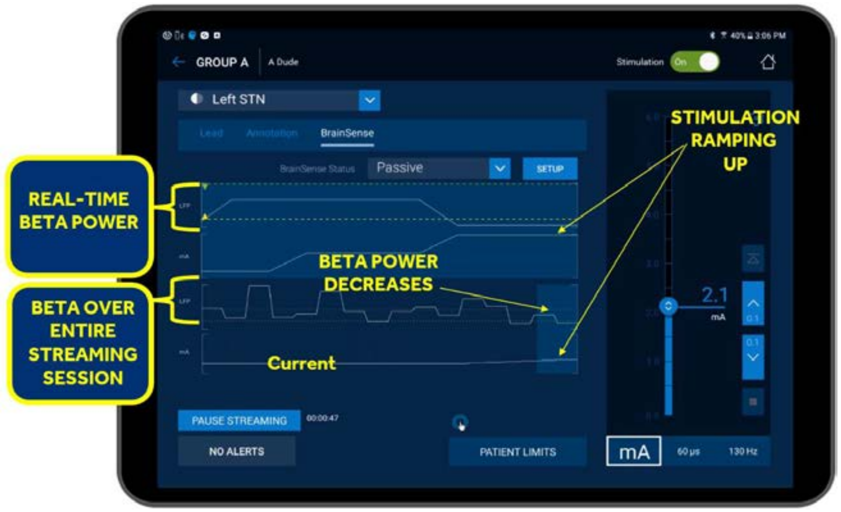
- When the LFP snapshot is captured, the neurostimulator measures approximately 30 seconds of LFP time domain data (250Hz sampling rate) and converts this to the frequency domain. Only the average frequency domain content is stored in the neurostimulator. Time domain data of the snapshot is **not** stored in the neurostimulator's memory. The 30 seconds of data is collected after the Patient Event is received by the neurostimulator.
- The frequency ranges from 0 to 96.68Hz. The upper frequency displayed on user interface may be limited, based on the stimulation frequency, to avoid displaying any potential stimulation artifacts from aliasing.



- Percept™ PC can record up to 400 LFP snapshots (e.g., 200 per hemisphere if bilateral). Percept™ RC can record up to 200 LFP snapshots (e.g., 100 per hemisphere if bilateral). If new snapshots are captured that exceed the maximal number of snapshots, then older snapshots are overwritten.
- Events that do not record LFP can also be captured by the patient. Percept™ PC can store up to 900 non-LFP events, while Percept™ RC can store up to 800 non-LFP events.

BrainSense™ Streaming

Purpose: Once BrainSense™ Setup has been completed, the user can view the LFP power in a selected frequency band in real time, by streaming the data to the clinician tablet. This is used to observe changes in the LFP during active stimulation programming or while instructing and observing the patient performing activities. Moreover, Streaming can be used to collect time domain data from the selected channel(s) for offline analysis and signal processing.



Used: In-clinic, to observe the LFP power in a frequency band over time, with or without stimulation.

Detailed Design:

- When start streaming is pressed, LFP data is streamed from both hemispheres, if both hemispheres have BrainSense™ Setup completed.
- Streaming data is viewed one lead at a time.
- The underlying sampling rate of the LFP power is 2Hz, but a unique solution is only calculated at the power averaging duration. Moreover, the power averaging duration is not a moving average it is a unique average, i.e. each average contains a unique set of data, not overlapping. This parameter is available in the Advanced Settings of the BrainSense™ Setup workflow. The underlying sampling of the time domain data is 250Hz.
- There is no maximum duration of streaming. However, consider that long streaming sessions:
  - Will fill the tablet memory sooner
  - Will take longer to generate the JSON. A610 version 4.0 supports exporting files with up to 8 hours of streaming data.
  - Long telemetry sessions can impact battery longevity for Percept™ PC devices, or recharge interval for Percept™ RC devices (see Longevity and Recharge Interval Considerations).
  - The tablet screen must be kept awake during streaming. For a Percept™ PC INS, user interaction is required to keep the screen awake (at least every 10 minutes if the screen timeout of the tablet operating system is left at the default 10-minute value). For a Percept RC INS, the clinician app will keep the screen awake for an additional hour while streaming, after which user interaction at least every 10 minutes is required.
- To optimize streaming performance and minimize the chance of dropping streaming data:
  - See the A610 Clinician Manual for Percept™ neurostimulators: M013075C001 for information about interference and how to best position the Communicator
  - Use shorter distances between the INS and Communicator. If not in an intra-operative scenario where the sterile field must be preserved, consider fixating the Communicator to the patient.
  - Minimize any obstacles or people between any components.
  - When streaming data is dropping, the BrainSense™ Streaming screen will show a gap in data. The JSON file has mechanisms such as sequence numbers, time 'tick' counters, and packet size information in addition to time stamp information in order to understand if data was dropped.

Note: There is also a "Record Streaming" option available via the BrainSense™ Survey feature that allows 6 time domain streams to be recorded – See the BrainSense™ Survey section for additional details.

Summary of Sensing Data

The data collected while using the BrainSense™ feature set is available in locations specified in Table 1 BrainSense™ Data Locations.

Table 1: BrainSense™ Data Locations

Sensing Feature Data	Time Domain [LFP Magnitude (μVp) vs. Time]	Frequency Domain [Magnitude (μVp) vs. Hz]	Power Domain [LFP Power in Band vs. Time]	Channels per Hemisphere (1x4 lead)	Channels per Hemisphere (SenSight™ lead)	Stim: On or Off	Data Collection: In or Out of Clinic
BrainSense Survey							
On-screen	-	X	-	6	15	Off	In
Session Report	-	X	-				
BrainSense Data Viewer	-	X	-				
JSON Export	X	X	-				
BrainSense Survey Indefinite Streaming							
On-screen	-	-	-	3	3	Off	In
Session Report	-	-	-				
BrainSense Data Viewer	-	-	-				
JSON Export	X	-	-				
BrainSense Setup (Signal Check)							
On-Screen	-	X	-	3	3	Both	In
Session Report	-	-	-				
BrainSense Data Viewer	-	X	-				
JSON Export	X	X	-				
BrainSense Timeline Data							
On Screen (timeline)	-	-	X	1	1	Either	Out
Session Report	-	-	Summarized				
BrainSense Data Viewer	-	-	X				
JSON Export	-	-	X				
Events (LFP Snapshot)							
On Screen	-	X	-	1	1	Either	Out
Session Report	-	-	-				
BrainSense Data Viewer	-	X	-				
JSON Export	-	X	-				
BrainSense Streaming							
On-Screen	-	-	X	1	1	Either	In
Session Report	-	-	-				
BrainSense Data Viewer	-	-	X				
JSON Export	X	-	X				

Advanced Topics

Longevity and Recharge Interval Considerations

See the System Eligibility and Battery Longevity manual (M929534A139) for the impact to Percept™ PC INS battery longevity due to chronically recording BrainSense™ signals (i.e. passive sensing).

Long clinician telemetry sessions with the Percept™ PC INS do have a small impact on the INS longevity. Using BrainSense™ streaming during these telemetry sessions does not add much additional energy usage since the primary energy use is the telemetry session itself. A rough order of magnitude estimate of the telemetry session impact for many patients is: a 1 hour telemetry session has approximately a 1 day impact to INS battery longevity.

For an implanted Percept™ RC neurostimulator, the recharge interval can be checked using the recharge interval calculator feature of the A610 application, which accounts for chronically recording BrainSense™ signals (i.e. passive sensing) if configured in the active group. See the A610 Clinician Programming Guide (M013075C001) for instructions.

Long clinician telemetry sessions (including using BrainSense™ streaming) will result in increased battery depletion on Percept™ RC devices and shorten the recharge interval. However, as the device is rechargeable, there is no impact on overall device longevity (service life). A rough order of magnitude estimate of the battery depletion for many patients is: a 1 hour streaming session will result in approximately 6% INS battery drain.

Signal Processing Parameters

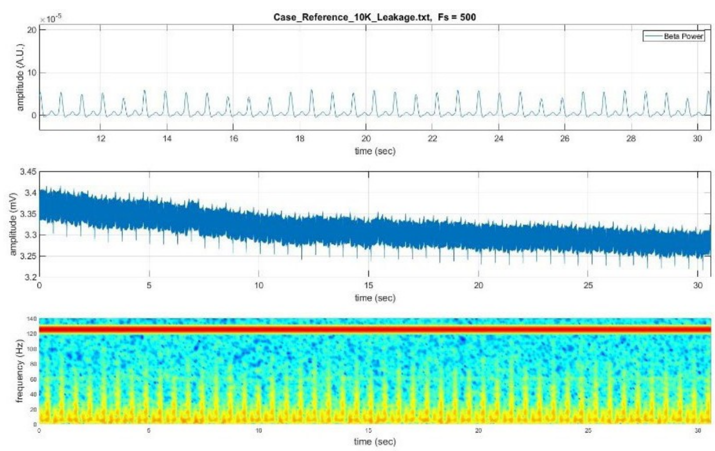
Some users may be interested in examining the raw measured LFP signal directly. For these users, it is important to understand how the raw data has been sampled and filtered. All recorded data is sampled at 250Hz. The data is passed through several filters. This includes 2 low pass filters at 100Hz, and two high pass filters. One high pass filter at 1Hz, and a second high pass filter at a user configurable 1Hz or 10Hz. The high pass filter setting is available in the Advanced Settings of the BrainSense™ Setup workflow.

Artifacts

An artifact is defined as a signal that is measured, which is not an LFP signal. The clinician application will display a message of artifact detected if it detects one of two possible artifact types; ECG and motion artifacts, discussed next. The system may also display artifact detected if the measured signal is significantly different than typical LFP signals. For example, a patient under anesthesia with only very large low frequency oscillations in LFP activity may be classified as artifact.

ECG Artifacts

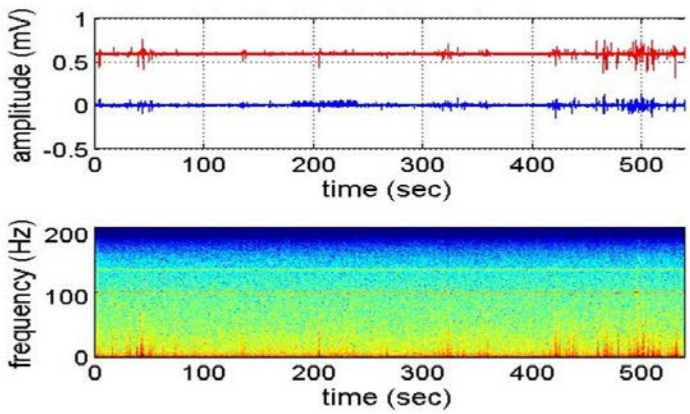
The first artifact the system detects is from the patient's heartbeat. It is possible that an ECG artifact conducts through the insulated pathway of the lead and extension. To investigate an ECG artifact, the time domain data in the .json file can be plotted (e.g. using Matlab). The example in Figure 1 is characteristic of an ECG artifact, where a large amplitude signal is observable at the heartrate of ~1-2Hz.



Medtronic Data on File: Figure 1: (Top) the power of the sensed signal in the selected band has an observable artifact at 1-2Hz. (Middle) the measured voltage of the raw data has a similar 1-2Hz artifact. (Bottom) the frequency spectrum has a broadband event at 1-2Hz.

Motion Artifacts

The second artifact the system detects is from the patient's movement. The artifactual signal in this scenario is not an LFP signal induced by motion, but instead a motion which causes the sensing apparatus (i.e. lead and/or extension) to move in a manner that produces an artifact. A motion artifact that is desired to be investigated is best observed when plotting the raw data in the .json file. The example in Figure 2 is characteristic of a motion artifact, where a very large amplitude signal is observable, that is not periodic, during times when the patient moves.



Medtronic Data on File: Figure 2: (Top) the measured voltage of the raw data has large amplitude spikes in it, that appear independent and not oscillatory. (Bottom) the frequency spectrum has broadband events during the spiking activity.

Artifact Override

The Artifact filter that prevents user selection of sensing channels that contain artifacts on BrainSense™ Setup and BrainSense™ Survey can be disabled in the Advanced Settings of the BrainSense™ Setup workflow.

Synchronization of data (external / internal)

Clinicians using external equipment such as accelerometers, video cameras, etc. may wish to synchronize these external devices with data from the Percept™ PC or RC device.

Coarse-grained synchronization (e.g. seconds or minutes resolution) can be performed by synchronizing the Percept™ PC or RC clock with the wall clock.

- Ensure that the tablet is on WIFI in order to sync the tablet clock with network time.
- Interrogate the INS
- Navigate to the "About" screen using the Navigation menu in the upper left corner
- Ensure the Time Source says "Tablet" and press the Update Device Time button

All data saved in the reports, including BrainSense™ data, is stored using UTC (ISO-8601) timestamps.

For fine-grained synchronization (e.g. sub-second resolution), using timing relative to some clinical event could be used. One potential method is as follows:

- Place EEG electrodes over INS and burr hole.
- Deliver low frequency (e.g. 50Hz) and low amplitude stimulation with the Percept™ device, which may be imperceptible to subject
- Start streaming with the Percept™ device and begin recording with external equipment.
- When complete with the activity, consider delivering a period of low frequency and/or low amplitude stimulation again with streaming on.
- Stimulation pulses can then be aligned between the raw LFP data (exportable via .json file) and external equipment using the low frequency/amplitude markers at the start/end of the streaming session
- For patients configured with stimulation cycling, the periodic on/off cycles in stimulation could be used for correlation when performing BrainSense™ Streaming.

JSON Export

The following high level data structures exist in the .json file that is exported when the user selects the “Export Json Session Data” option from the Reports functionality of the Clinician Programmer. Once generated the .json file is stored on the tablet in the /Reports folder, along with any generated session reports, MRI reports, etc.

Session Data and the JSON Export

The JSON export contains a summary of information in JSON format, rendered to be machine readable. The DBS Clinician app provides options to download a PDF of the Session report from each therapy session, or to export the same data in JSON format.

The Json Export Session Data report contains the following information:

- Patient Information
- Device Information
- Battery Information
- Group Usage Percentage since last session in the form of combined groups array
- Lead Configuration
- Stimulation
- Groups
- Battery Check Reminder
- Battery Recharge (for applicable devices, currently Activa RC and Percept™ RC)
- Recharge Interval estimation (Percept™ RC Only)
- Non-Recharge Battery Status (Percept™ RC Only)
- Impedance
- Group History
- Annotations
- LFP Montage (Percept™ devices Only)
- Patient Events (Percept™ devices Only)
- Event Summary (Percept™ devices Only)
- Diagnostics Data
- BrainSense Streaming
- Indefinite Streaming

The JSON Export Session Data report is generated using the app’s Patient Data Service.

JSON Structure Format

The information is fetched from various features in the app using respective mechanisms to retrieve the device state at the beginning/end of each therapy session. The information shown in the export varies, depending on whether all logs have been read from the device and stored or if activities like impedance measurements or streaming were conducted during the session. The JSON export file will indicate if the information in the export is incomplete, such as would be the case in events preventing a complete device reading of all logs. An example is a user canceling the background loading process.

Relevant information is grouped together as JSON Objects or Arrays in the output file.

Enumerated types are not translated as per JSON Normalization for Translation. Instead the enumeration is rendered as a _Def.name such that a consumer of the JSON can translate the enumeration to the proper locale using the JSON Enum Translation Files.

All the times mentioned in the export are UTC (ISO 8601) only. In addition, programmer locale and time zone values are included in the export.

All fields shown in the Session report need to be shown in the JSON export as well. In addition to field values, there are three types of raw data in the JSON export: Time Domain, Thresholds, and BrainSense™ LFP. Please see below for explanations of all JSON data structures and data sub-structures.

MATLAB Import

For Matlab import of .json files, see Matlab help documentation for command jsondecode. This command exists in Matlab version 2016b or later. The image below shows the Matlab workspace after importing a .json file using *jsondecode*. The data exists as a hierarchy of data structures.

Field	Value
AbnormalEnd	0
FullyReadForSession	1
FeatureInformationCode	'1X6'
SessionDate	'2019-10-04T20:08:29Z'
SessionEndDate	'2019-10-04T20:32:12Z'
ProgrammerTimezone	'Central Daylight Time'
ProgrammerUtcOffset	'-05:00'
ProgrammerLocale	'en_US'
ProgrammerVersion	'2.0.4580'
PatientInformation	*1x1 struct*
DeviceInformation	*1x1 struct*
BatteryInformation	*1x1 struct*
GroupUsagePercentage	*4x1 cell*
LeadConfiguration	*1x1 struct*
Stimulation	*1x1 struct*
Groups	*1x1 struct*
BatteryReminder	*1x1 struct*
MostRecentInSessionSignalCheck	*6x1 struct*
Impedance	*1x1 struct*
GroupHistory	*2x1 struct*
SenseChannelTests	*6x1 struct*
CalibrationTests	*4x1 struct*
Thresholds	*6x1 struct*
LfpMontageTimeDomain	*12x1 struct*
IndefiniteStreaming	*6x1 struct*
BrainSenseTimeDomain	*3x1 struct*
BrainSenseLfp	*2x1 struct*
LFPMontage	*12x1 struct*
PatientEvents	*1x1 struct*
EventSummary	*1x1 struct*
DiagnosticData	*1x1 struct*

One example of matlab code used for loading a .json file is:

[filename, pathname] = uigetfile('*.json'); json

= fileread([pathname,'\',filename]); datastruct

= jsondecode(json);

Viewing .json files

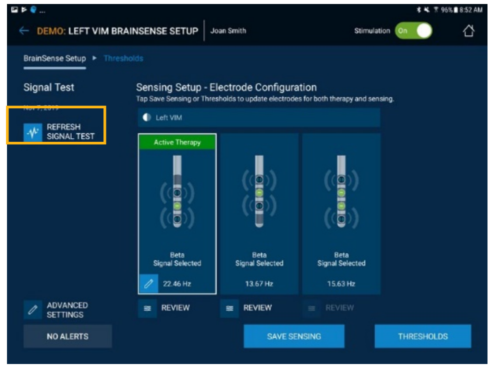
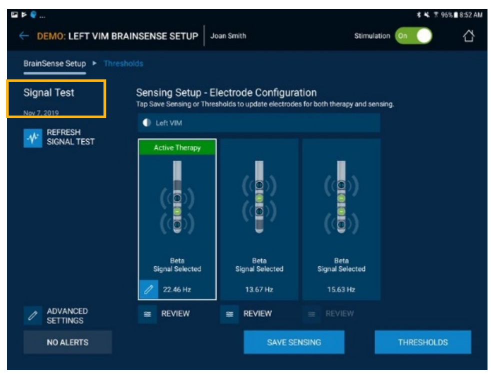
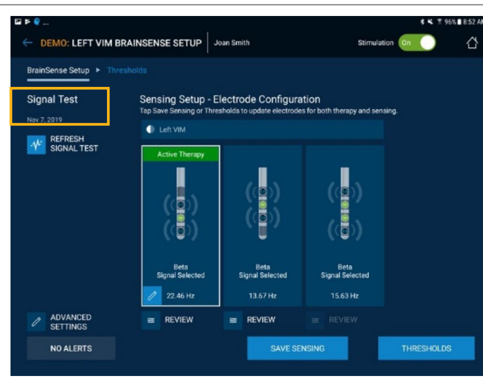
To view .json files, they can be opened in any text editor. For example, install Notepad++ and open the .json file. The image below is a Notepad++ view of an open .json file. Specific to Notepad++ the JSTool Plugin can be used to better navigate the .json file with a navigation pane. Several other open-source tools can be used to view .json files.

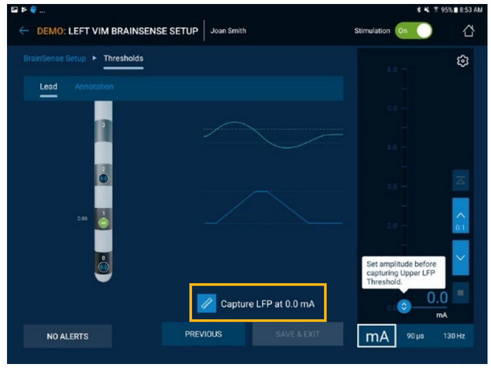
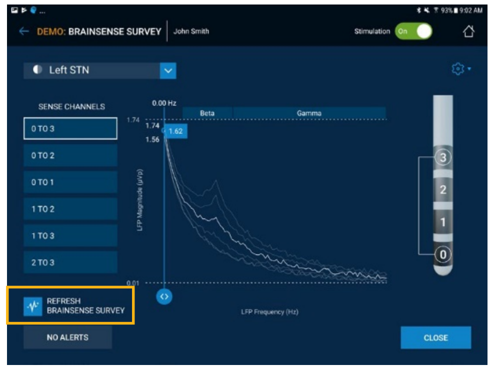
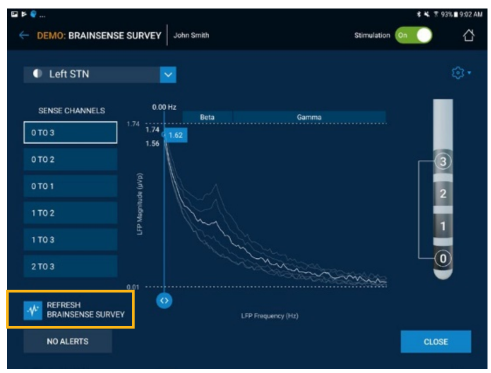
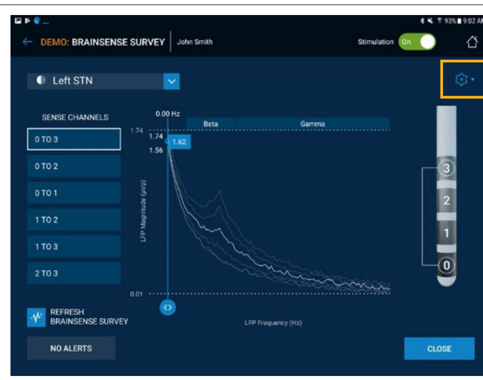
```
    "LeadConfiguration": {
        "Initial": [
            {
                "Hemisphere": "HemisphereLocationDef.Left",
                "Model": "LeadModelDef.LEAD_3387",
                "LeadLocation": "LeadLocationDef.Stn",
                "ElectrodeNumber": "InsPort.ZERO_THREE",
                "TipOffset": "TipOffsetDef.ZERO"
            },
            {
                "Hemisphere": "HemisphereLocationDef.Right",
                "Model": "LeadModelDef.LEAD_3387",
                "LeadLocation": "LeadLocationDef.Stn",
                "ElectrodeNumber": "InsPort.EIGHT_ELEVEN",
                "TipOffset": "TipOffsetDef.EIGHT"
            }
```

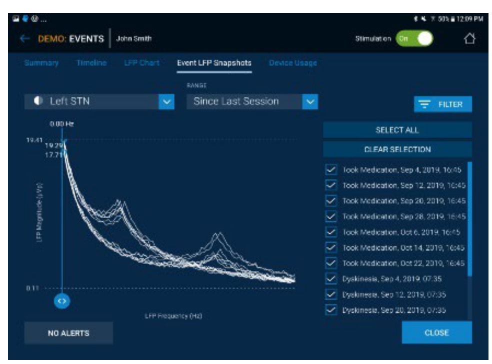
Note that some structures contain an 'initial' and 'final' state (where applicable) to highlight changes made during that session.

Sensing Specific Data Structures

Table 2: Sensing Specific Data Structures

Data Name	Description	Screen User uses to Collect Data	# of Records when used multiple times in a single CP session
MostRecentInSessionSignalCheck	This section contains the last sense signal quality check results that session (3 channels amp/freq per lead) – stimulation is OFF Whenever the application streams time domain data for the quality check, the data is written to a file in the Patient Data Service (PDS) for later use. The clinician software performs a quality check for each set of LFP measurements. Then the clinician software uses the signal quality check algorithm to confirm there are no artifacts. The algorithm looks for: • Cardiac artifacts • Motion artifacts If artifacts are detected on a sense channel, the user is informed and allowed to retry the check.		1, the most recent only.
SenseChannelTests	This section contains the time domain data used to compute the sense signal quality result with stim OFF (3 channels per lead) The application runs a signal check when all sensing preconditions have been met. If the preconditions for sensing are not met, the button will be disabled and the user will not be allowed to run a signal check.		All times performed that session.
CalibrationTests	This section contains the time domain data used to compute the sense signal quality result with stim ON (2 channels per lead)		All times performed that session.

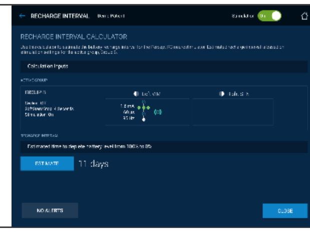
Thresholds	This section contains the power domain data used to compute any sensing thresholds set in this session		All times performed that session.
LfpMontageTimeDomain	This section contains the time domain data used to compute the freq/amp plot in BrainsenseSurvey (6 channels per 1x4 lead, up to 15 channels per SenSight™ lead) – stimulation is OFF		All times performed that session.
LFPMontage	This section contains the LFP amplitude and frequency data displayed to user from BrainsenseSurvey (6 channels per 1x4 lead, up to 15 channels per SenSight™ lead) – stimulation is OFF		1, the most recent only.
IndefiniteStreaming	This section contains the time domain data recorded when using the “Record Streaming” option from BrainSenseSurvey Gear icon (3 channels per lead)– stimulation is off.		All times performed that session.

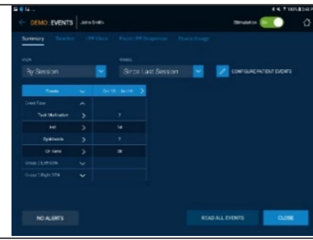
BrainSenseTimeDomain	This section contains time domain data gathered when the user presses 'Start streaming" in BrainSense™ workflow – stimulation is in user defined state.		All times performed that session.
BrainSenseLfp	This section contains the raw data (Power data) gathered when the user presses 'Start streaming" in BrainSense™ workflow – stimulation is in user defined state.		All times performed that session.
DiagnosticData	This section contains the LFP Trend (60 days of LFP and Stim Amplitude on Percept™ PC, 35 days of LFP and Stim amplitude on Percept™ RC; 10 min average data), the Device Event Logs (e.g., change group, stimulation on/off, clinician session start/end, impedance measurement) and the LFP Frequency Snapshot Events raw data (time of event, and raw LFP data (Freq/Amp)) for each event. All diagnostic data since the last session is saved on the tablet in the PDS session report, unless a request is made for a 'full' report, which would contain all diagnostic data on the INS. This data saved in PDS is available via a JSON export.	 & 	N/A, not a task that can be performed multiple times in a single CP session.

Other Data Structures

Table 3: Other Data Structures

Data Structure Name	Description/Contents	Screen User uses to Collect Data	# of Records when used multiple times in a single CP session
AbnormalEnd	This section is to indicate whether the clinician session for which the report is generated was ended abnormally. Possible causes for abnormal session end can be an application crash or an event where the application is killed from background, etc.	N/A: Data is read automatically at Device interrogation.	N/A, not a task that can be performed multiple times in a single CP session.
FullyReadForSession	When connecting to an INS, the application first performs an initial series of reads, checks, and validations before the user gets to the home screen. Once a user is at the home screen, the CP App will continue to load historical data in the background (historical data includes timeline data, sensing data, patient marked events, etc. that were collected since the prior CP session). This field indicates if that background loading was completed during the session (marked 'true') or if the data was not fully read during the session (marked 'false'). If false, some structures may be missing from the JSON file.	N/A: Data is read automatically at Device interrogation.	N/A, not a task that can be performed multiple times in a single CP session.
FeatureInformationCode	This code corresponds to the features the software permits to be enabled based on device type and/or the region in which the device is being utilized.	N/A: Data is read automatically at Device interrogation.	N/A, not a task that can be performed multiple times in a single CP session.
SessionDate	The system stamps each session with a time and date upon initiation. This includes the month, day, and year. The time is in 24-hour format and includes hours, minutes, and seconds. The format varies between locales.	N/A: Data is read automatically at Device interrogation.	N/A, not a task that can be performed multiple times in a single CP session.
SessionEndDate	The system stamps each session with an end date upon session end. It includes the same information as the session date. The end time is maintained exclusively for the JSON export data.	N/A: Data is read automatically at Device interrogation.	N/A, not a task that can be performed multiple times in a single CP session.
ProgrammerTimezone	The time zone in which the programmer is being used. Ex. Central Daylight Time.	N/A: Data is read automatically at Device interrogation.	N/A, not a task that can be performed multiple times in a single CP session.
ProgrammerUtcOffset	This is the value needed to adjust the UTC time value to match the selected time zone.	N/A: Data is read automatically at Device interrogation.	N/A, not a task that can be performed multiple times in a single CP session.
ProgrammerLocale	This is the locale in which the user selects to operate the programmer. Aside from English, there are 27 language options.	N/A: Data is read automatically at Device interrogation.	N/A, not a task that can be performed multiple times in a single CP session.
ProgrammerVersion	The version of the programmer in use. In "0.0.0" format.	N/A: Data is read automatically at Device interrogation.	N/A, not a task that can be performed multiple times in a single CP session.
PatientInformation	This section contains name/gender/DOB/ ID/Notes/Diagnosis – can be de-identified when generated	N/A: Data is entered by the user in Setup workflow and read automatically at Device interrogation.	Contains an initial (start of session) and final (end of session) record.
DeviceInformation	This section contains INS Model/ SN/Version/Implant Locations/ AccumulatedTherapyOnTime	N/A: Data is read automatically at Device interrogation.	Contains an initial (start of session) and final (end of session) record.

BatteryInformation	This section contains Percentage and additional information such as estimated remaining battery life (Percept™ PC) or ERI date (Percept™ RC)	N/A: Data is read automatically at Device interrogation.	N/A, not a task that can be performed multiple times in a single CP session.
BatteryReminder	This section contains the status and time of the patient check battery reminder, if enabled.	N/A: Data is read automatically at Device interrogation	N/A, not a task that can be performed multiple times in a single CP session.
RechargeCount	This section contains dates, durations, voltages, and other information about recent INS recharge sessions. This section is only available for rechargeable INS models	N/A: Data is read automatically at Device interrogation.	N/A, not a task that can be performed multiple times in a single CP session.
RechargeInterval	This section contains the results of the Recharge Interval Calculator tool. This section is only available for Percept™ RC.		1, the most recent only.
NonRechargeBatteryStatus	This section contains non-recharging information about the battery of a connected Percept™ RC device. This includes battery percentages, the battery state, and a recent battery voltage measurement. This section is only available for Percept™ RC	N/A: Data is read automatically at Device interrogation.	N/A, not a task that can be performed multiple times in a single CP session.
GroupUsagePercentage	This section contains % of time in each group	N/A: Data is read automatically at Device interrogation.	N/A, not a task that can be performed multiple times in a single CP session.
LeadConfiguration	This section contains lead models, implant locations, and the lead orientation value for SenSight™ leads	N/A: Data is read automatically at Device interrogation.	Contains an initial (start of session) and final (end of session) record.
Stimulation	This section contains stim ON/OFF status at start and end of session	N/A: Data is read automatically at Device interrogation.	Contains an initial (start of session) and final (end of session) record.
Groups	This section contains all programmable stimulation and sensing settings for every group configured.	N/A: Data is read automatically at Device interrogation.	Contains an initial (start of session) and final (end of session) record.
Annotations	This section contains the dates, electrode configurations, side effects, and symptoms for saved annotations.		Contains each annotation entered during (and prior to) the session.
Impedance	This section contains electrode impedance results (in ohms for each electrode pair) run that session		1, the most recent only.
GroupHistory	This section contains the session history stored on the device (up to last 5 Clinician Programmer sessions)	N/A: Data is read automatically at Device interrogation.	N/A, not a task that can be performed multiple times in a single CP session.

PatientEvents	This section contains the names of the Events for the patient, and their additional behavior (i.e. take LFP / restart cycling)	N/A: Data is read automatically at Device interrogation.	Contains an initial (start of session) and final (end of session) record.
EventSummary	This section contains the data used to populate the table on summary screen in CP app (above/below/between threshold percentage for each group – if sensing) and event counts used since last session.		N/A, not a task that can be performed multiple times in a single CP session.

Data Structure Details and Substructures

For each of the high level data structures in Table 2: Sensing Specific Data Structures and Table 3: Other Data Structures, Table 4: Data Structure Details and Substructures details the substructure (lower level) data within the high level structure.

Table 4: Data Structure Details and Substructures

Data Structure Name	SubStructures	Units and Sample Rate
MostRecentInSessionSignalCheck	6 Sensing Channels (3 electrode level pairs per lead)	N/A
	Each Channel Contains the following:	
	SignalFrequencies	Units of Hz. Post Spectral Analysis
	SignalPsdValues	Units of µVp. Post Spectral Analysis
SenseChannelTests	6 Sensing Channels (3 electrode level pairs per lead) Contains all Signal Quality Check data if run multiple times.	N/A
	Each Channel Contains the following:	
	SampleRateInHz	Units of Hz.
	TimeDomainData	Units of µV, with sample rate of SampleRateInHz
	Global Sequences	Lists packet identity sequentially. Used to identify missing packets, by missing sequence numbers. Rolls over at 255 for Percept™ PC or 65,535 for Percept™ RC.
	GlobalPacketSizes	Lists number of samples per packet
	FirstPacketDatetime	Labels each Streaming Sample with initial DateTime value. 1 second resolution
CalibrationTests	4 Sensing Channels (2 electrode level pairs per lead) Contains all Calibration test data if run multiple times. Calibration Test is the 2nd step of signal quality checking, performed with stimulation on.	N/A
	Each Channel Contains the following:	
	SampleRateInHz	Units of Hz.
	TimeDomainData	Units of µV, with sample rate of SampleRateInHz
	Global Sequences	Lists packet identity sequentially. Used to identify missing packets, by missing sequence numbers. Rolls over at 255 for Percept™ PC or 65,535 for Percept™ RC.
	GlobalPacketSizes	Lists number of samples per packet
	FirstPacketDatetime	Labels each Streaming Sample with initial DateTime value. 1 second resolution
Thresholds	Each Threshold measurement Contains the hemisphere it was streamed from, and the following substructures:	N/A
	SampleRateInHz	Units of Hz
	LFPData: One data structure per hemisphere (left/right)	This number represents the LFP power of the selected frequency band. It is calculated as the sum of the squared LFP magnitude at each frequency within the selected band, similar to the Area Under the Curve (AUC), and has a sample rate of SampleRateInHz.

LfpMontageTimeDomain	12 Sensing Channels for two 1x4 leads (6 electrode pairs per lead), all user requested measurements from the session are saved, 2 surveys (1 per lead) needed to capture all available sensing channels First survey contains all electrode pair combinations for first hemisphere. Second survey contains all electrode pair combinations for second hemisphere. 30 Sensing Channels for two SenSight™ leads (15 electrode pairs per lead), all user requested measurements from the session are saved, 4 surveys (2 per lead) needed to capture all available sensing channels First survey contains all electrode level pair combinations for first hemisphere. Second survey contains all segment-to-segment pair combinations in two passes for first hemisphere. Third survey contains all electrode level pair combinations for second hemisphere. Fourth survey contains all segment-to-segment pair combinations in two passes for second hemisphere.	N/A
	Each Channel Contains the following:	
	SampleRateInHz	Units of Hz
	TimeDomainData	Units of µV, with sample rate of SampleRateInHz
	Global Sequences	Lists packet identity sequentially. Used to identify missing packets, by missing sequence numbers. Rolls over at 255 for Percept™ PC or 65,535 for Percept™ RC.
	GlobalPacketSizes	Lists number of samples per packet
	FirstPacketDatetime	Labels each Streaming Sample with initial DateTime value. 1 second resolution
LFPMontage	12 Sensing Channels for two 1x4 leads (6 electrode pairs per lead) 30 Sensing Channels for two SenSight™ leads (15 electrode pairs per lead)	N/A
	Each Channel Contains the following:	
	ArtifactStatus	Flag for detecting artifact
	LFPFrequency	Units of Hz. Post Spectral Analysis
	LFPMagnitude	Units of µVp. Post Spectral Analysis

IndefiniteStreaming	6 Sensing Channels (3 electrode level pairs per lead). Contains non-adjacent electrode configurations.	N/A
	SampleRateInHz	Units of Hz
	TimeDomainData	Units of µV, with sample rate of SampleRateInHz
	Global Sequences	Lists packet identity sequentially. Used to identify missing packets, by missing sequence numbers. Rolls over at 255 for Percept™ PC or 65,535 for Percept™ RC.
	GlobalPacketSizes	Lists number of samples per packet
	TicksInMses	Lists the system time tick for each packet sent. 50 millisecond resolution
	FirstPacketDatetime	Labels each Streaming Sample with initial DateTime value. 1 second resolution
BrainSenseTimeDomain	Each Streaming Sample Contains the sensing channel the data is from, (the hemisphere and electrodes), and then has the following substructures:	N/A
	SampleRateInHz	Units of Hz
	TimeDomainData	Units of µV, with sample rate of SampleRateInHz
	Global Sequences	Lists packet identity sequentially. Used to identify missing packets, by missing sequence numbers. Rolls over at 255 for Percept™ PC or 65,535 for Percept™ RC. Packets are interleaved with BrainSenseLfp packets.
	GlobalPacketSizes	Lists number of samples per packet
	TicksInMses	Lists the system time tick for each packet sent. 50 millisecond resolution.
	FirstPacketDatetime	Labels each Streaming Sample with initial DateTime value. 1 second resolution

BrainSenseLfp	Each Streaming Sample Contains the sensing channel the data is from, (the hemisphere and electrodes), and then has the following substructures:	N/A
	SampleRateInHz	Units in Hz.
	TherapySnapshot	Group/Program settings specific to streaming sample
	LfpData: One data structure per hemisphere (left/right)	LFP: This number represents the LFP power of the selected frequency band. It is calculated as the sum of the squared LFP magnitude at each frequency within the selected band, similar to the Area Under the Curve (AUC), and has a sample rate of SampleRateInHz. mA: Units of mA, with sample rate of SampleRateInHz. This is the Stimulation amplitude.
	FirstPacketDateTime	Labels each Streaming Sample with initial DateTime value. 1 second resolution
	Seq	Labels each LFPData sample with packet identity. Used to identify missing packets, by missing sequence numbers. Rolls over at 255 for Percept™ PC or 65,535 for Percept™ RC. Packets are interleaved with BrainSenseTimeDomain GlobalSequences.
	TicksInMs	Labels each LfpData sample with the system time tick. 50 millisecond resolution. Note: TicksInMs normally roll over every 2^16 ticks, or once every 54.61 minutes (65535 ticks * 50 ms per tick = 3,276,750ms). However, during streaming, TicksInMs will **NOT** roll over and will continue increasing until streaming is stopped.
DiagnosticData	Contains various Logs for tracking outside of clinician visit	N/A
	On Percept™ PC: LfpTrendLogs: Contains up to 60 days of 10 minute averaged LFP and stim amplitude per hemisphere. *On Percept™ RC:* LfpTrendLogs: Contains up to 35 days of 10 minute averaged LFP and stim amplitude per hemisphere.	DateTime: Time of sample LFP: This number represents the LFP power of the selected frequency band. It is calculated as the sum of the squared LFP magnitude at each frequency within the selected band, similar to the Area Under the Curve (AUC). AmplitudeInMilliAmps: Units of mA. Stim amplitude Data may be censored to avoid artifacts, censored data is negative.
	EventLogs	Patient Interaction Log: (e.g. Switching groups, session start, stim on/off, Impedance Test)

	On Percept™ PC: LfpFrequencySnapshotEvents: Contains up to 400 LFP Frequency snapshots, 200 if 2 hemispheres. *On Percept™ RC:* LfpFrequencySnapshotEvents: Contains up to 200 LFP Frequency snapshots, 100 if 2 hemispheres. *On either Percept™ PC or RC:* Use “Read All Events” button in clinician application before exporting .json file to load all Snapshot Events. 2 substructures, 1 per hemisphere	Frequency: Units in Hz, Post Spectral Analysis FFTBinData: Units in µVp, Post Spectral Analysis Data may be censored to avoid artifacts, censored data is negative.
PatientInformation	Contains text for patient first and last name, gender, DOB, ID, ClincianNotes, and Diagnosis.	N/A

DeviceInformation	Contains INS Model, Serial Number, Firmware Versions, Implant Locations and AccumulativeTherapyONTimeSinceImplant/Followup	TherapyOnTime is in seconds
BatteryInformation	Contains Battery Percentage, Estimated Battery Life Remaining (in months) for Percept™ PC, ERI Date and average recharge coupling for Percept™ RC, Battery Status	Percent is out of 100, and battery life is months.
BatteryReminder	Contains the status and the reminder time of the Patient Check Battery Reminder, if enabled	The reminder time is saved as the hour and minute of the notification.
RechargeCount	Various substructures for each recent recharge event. Each substructure contains a recharge ID, recharge duration, recharge coupling, recharge session date, as well as the starting/ending voltages, and battery percentages. Note: Information available may vary depending on rechargeable INS model	Timestamps have 1 second resolutions. Voltages are in millivolts. Battery percentages are out of 100
RechargeInterval	Contains the estimated recharge interval (Percept RC only), as well as a substructure of calculation inputs used for the estimate of the recharge interval calculation.	The recharge estimate is in days
NonRechargeBatteryStatus	Contains Battery level percentage, states of charge (and uncertainty), battery state, and voltage measurements	Battery Level Percentage is out of 100. Voltages are in millivolts.
GroupUsagePercentage	4 substructures, one for each of Group A,B,C,D. Each contains usage percentage for that Group	Percent is out of 100
LeadConfiguration	2 substructures, one for each lead (one lead per hemisphere). Contains the lead model, the lead location, the INS bore/electrodes numbers, the tip electrode number for each lead, and the lead orientation value in degrees for SenSight™ leads.	N/A
Stimulation	Contains the Stimulation Status ON or OFF.	N/A
Groups	Initial and Final structures contain up to 4 substructures, one for each Group. Within each substructure is the Group Name, if it is the Active Group or not, the patient adjustable stimulation parameter for that group, and the following two substructures:	N/A
	ProgramSettings – contains program level stimulation and sensing parameters. Amplitude per electrode, Rate, Pulse Width, Electrodes, Thresholds, Sensing Frequency, Averaging Duration, Sense Channel Result for the Active channel.	Units are contained in variable name
	GroupSettings – contains group level stimulation and sensing parameters. SoftStart, Cycling, highpassfilter, sense blanking duration.	Units are contained in variable name
Annotations	Contains substructure for each annotation. Annotations contain dates, stimulation frequencies and the hemisphere of interest, in addition to Program Settings, Symptoms, and Side Effects.	Program settings units are contained in variable name
Impedance	Contains electrode impedance results for impedance test during that session. 2 substructures one for each hemisphere (lead). Within each substructure, 2 additional substructures:	N/A
	Monopolar: Contains the monopolar electrode pair impedance results	Ohms
	Bipolar: Contains the bipolar electrode pair impedance results	Ohms

GroupHistory	Contains up to 5 Group histories from 5 previous clinician programming sessions on different days. Within each of the 5 previous programming days, up to 4 substructures exist, one for each group. These substructures follow the same format as the Groups data structure above.	N/A
PatientEvents	Contains the event name of up to four events, and if each event has additional behavior (LFP Snapshot or Restart Cycling).	N/A
EventSummary	Contains the data used to populate the table on summary screen in CP app (above/below/between threshold percentage for each group and average stimulation amplitude – if sensing) and event count used since last session.	Percent is out of 100. Average amplitude is in mA.

Time Domain Data Point Timestamps

BrainSenseTimeDomainData and IndefiniteStreaming have TimeDomainData structures with a clock value per telemetry message. Each message contains a variable number of data points. The following pseudocode is an example method to estimate the relative time each data point was sampled.

FOR each streaming sample:

1. Create time array [e.g. TDtime] for the TimeDomainData contained in the last (most recent) telemetry message.
2. Assign the last TimeDomainData data point to the last TicksInMses value.
3. Iterate backward in time over the size (GlobalPacketSizes) of the last message received, knowing that each data point in the message is spaced at 1/ SampleRateInHz.

   For example:
   TDtime = TicksInMses (end)- (GlobalPacketSizes (end)-1)/SampleRateInHz : 1/SampleRateInHz : TicksInMses (end)

FOR each prior message, starting from the last message above

IF Difference in TicksInMses > (1 + GlobalPacketSizes)/ SampleRateInHz

1. Assign the next most recent TimeDomainData data point to the next most recent TicksInMses value
2. Iterate backward in time over the size (GlobalPacketSizes) of the next most recent message received.
3. Assemble the data vector by appending the messages:

   [Next Most Recent, Most Recent]

   For example:
   Prev_packet = TicksInMses (i-1)- (GlobalPacketSizes (i-1)-1)/ SampleRateInHz :… 1/SampleRateInHz : TicksInMses (i-1TDtime = [Prev_packet,TDtime]

ELSE

1. Assign the next most recent TimeDomainData data point to 1/SampleRateInHz prior to the time value computed for the first data point in the prior message, i.e. assume the data is continuous.
2. Iterate backward in time over the size (GlobalPacketSizes) of the next most recent message received.
3. Assemble the data vector by appending the messages:

   [Next Most Recent, Most Recent]

   For example:
   Prev_packet = TDtime(1)- GlobalPacketSizes (i-1)/ SampleRateInHz: 1/SampleRateInHz :
   TDtime(1) – 1/SampleRateInHzTDtime = [Prev_packet,TDtime]

END

Optionally, use the first value in variable TicksInMses as a zero reference for plotting, by subtracting this value from the time vector.

Glossary

Abbreviation	Definition
CP App	Clinician Programmer Application
LFP	Local Field Potential
PD	Parkinson’s Disease
UTC	Coordinated Universal Time (ISO-8601)
µVp	Micro volts peak

Medtronic's DBS Therapy is approved for 5 indications: Parkinson's disease, essential tremor, dystonia*, obsessive-compulsive disorder* (OCD), and epilepsy. Device indications vary, refer to product labeling.

Medtronic did not seek FDA approval for the SenSight™ or Percept™ RC systems for OCD. The updated A610 version 4.0 clinician programmer application and updated A620 version 3.0 patient programmer application will support OCD programming for patients implanted with Activa™/Percept™ PC neurostimulation systems, including the Model 3391 lead system.

***Humanitarian Device:** The effectiveness of these devices for the treatment of dystonia and obsessive-compulsive disorder has not been demonstrated.

Brief Statement: Medtronic DBS Therapy for Parkinson's Disease, Tremor, Dystonia, Obsessive-Compulsive Disorder, and Epilepsy

Product labeling must be reviewed prior to use for detailed disclosure of risks.

INDICATIONS:
Medtronic DBS Therapy for Parkinson's Disease: Bilateral stimulation of the internal globus pallidus (GPi) or the subthalamic nucleus (STN) using Medtronic DBS Therapy for Parkinson's Disease is indicated for adjunctive therapy in reducing some of the symptoms in individuals with levodopa-responsive Parkinson's disease of at least 4 years' duration that are not adequately controlled with medication, including motor complications of recent onset (from 4 months to 3 years) or motor complications of longer-standing duration.

Medtronic DBS Therapy for Tremor: Unilateral thalamic stimulation of the ventral intermediate nucleus (VIM) using Medtronic DBS Therapy for Tremor is indicated for the suppression of tremor in the upper extremity. The system is intended for use in patients who are diagnosed with essential tremor or parkinsonian tremor not adequately controlled by medications and where the tremor constitutes a significant functional disability.

Medtronic DBS Therapy for Dystonia*: Unilateral or bilateral stimulation of the internal globus pallidus (GPi) or the subthalamic nucleus (STN) using Medtronic DBS Therapy for Dystonia is indicated as an aid in the management of chronic, intractable (drug refractory) primary dystonia, including generalized and/or segmental dystonia, hemidystonia, and cervical dystonia (torticollis), in patients seven years of age or above.

Medtronic DBS Therapy for Obsessive-Compulsive Disorder*: The Medtronic Reclaim™ DBS Therapy is indicated for bilateral stimulation of the anterior limb of the internal capsule, AIC, as an adjunct to medications and as an alternative to anterior capsulotomy for treatment of chronic, severe, treatment-resistant obsessive-compulsive disorder (OCD) in adult patients who have failed at least three selective serotonin reuptake inhibitors (SSRIs).

Medtronic DBS Therapy for Epilepsy: Bilateral stimulation of the anterior nucleus of the thalamus (ANT) using the Medtronic DBS System for Epilepsy is indicated as an adjunctive therapy for reducing the frequency of seizures in individuals 18 years of age or older diagnosed with epilepsy characterized by partial-onset seizures, with or without secondary generalization, that are refractory to three or more antiepileptic medications.

The Medtronic DBS System for Epilepsy has demonstrated safety and effectiveness for patients who average six or more seizures per month over the three most recent months prior to implant of the DBS system (with no more than 30 days between seizures). The Medtronic DBS System for Epilepsy has not been evaluated in patients with less frequent seizures.

CONTRAINDICATIONS: Medtronic DBS Therapy is contraindicated for patients who are unable to properly operate the neurostimulator and, for Parkinson's disease and Essential Tremor, patients for whom test stimulation is unsuccessful. The following procedures are contraindicated for patients with DBS systems: diathermy (e.g., shortwave diathermy, microwave diathermy or therapeutic ultrasound diathermy), which can cause neurostimulation system or tissue damage and can result in severe injury or death; Transcranial Magnetic Stimulation (TMS); and certain MRI procedures using a full body transmit radio-frequency (RF) coil, a receive-only head coil, or a head transmit coil that extends over the chest area if they have an implanted Soletra™ Model 7426 Neurostimulator, Kinetra™ Model 7428 Neurostimulator, Activa™ SC Model 37602 Neurostimulator, or Model 64001 or 64002 pocket adaptor.

WARNINGS: There is a potential risk of brain tissue damage using stimulation parameter settings of high amplitudes and wide pulse widths and, for Parkinson's disease and Essential Tremor, a potential risk to drive tremor (cause tremor to occur at the same frequency as the programmed frequency) using low frequency settings. Extreme care should be used with lead implantation in patients with an increased risk of intracranial hemorrhage. Sources of electromagnetic interference (EMI) may cause device damage or patient injury. Theft detectors and security screening devices may cause stimulation to switch ON or OFF and may cause some patients to experience a momentary increase in perceived stimulation. The DBS System may be affected by or adversely affect medical equipment such as implanted cardiac devices (e.g., pacemaker, defibrillator), external defibrillation/cardioversion, ultrasonic equipment, electrocautery, or radiation therapy. MRI conditions that may cause excessive heating at the lead electrodes which can result in serious and permanent injury including coma, paralysis, or death, or that may cause device damage, include: neurostimulator implant location other than pectoral and abdominal regions; unapproved MRI parameters; partial system explants ("abandoned systems"); misidentification of neurostimulator model numbers; and broken conductor wires (in the lead, extension or pocket adaptor). The safety of electroconvulsive therapy (ECT) in patients receiving DBS Therapy has not been established. Abrupt cessation of stimulation should be avoided as it may cause a return of disease symptoms, in some cases with intensity greater than was experienced prior to system implant ("rebound" effect). Onset of status dystonicus, which may be life-threatening, may occur in dystonia patients during ongoing or loss of DBS therapy.

For Epilepsy, cessation, reduction, or initiation of stimulation may potentially lead to an increase in seizure frequency, severity, and new types of seizures. For Epilepsy, symptoms may return with an intensity greater than was experienced prior to system implant, including the potential for status epilepticus. For Parkinson's disease or essential tremor, new onset or worsening depression, suicidal ideation, suicide attempts, and suicide have been reported. For Dystonia or Epilepsy, depression, suicidal ideations and suicide have been reported, although no direct cause-and-effect relationship has been established. For Epilepsy, preoperatively, assess patients for depression and carefully balance this risk with the potential clinical benefit. Postoperatively, monitor patients closely for new or changing symptoms of depression and manage these systems appropriately. Patients should be monitored for memory impairment. Memory impairment has been reported in patients receiving Medtronic DBS Therapy for Epilepsy, although no direct-cause-and effect relationship has been established. The consequences of failing to monitor patients are unknown. When stimulation is adjusted, monitor patients for new or increased seizures, tingling sensation, change in mood, or confusion. For Obsessive-Compulsive Disorder, patients should be monitored for at least 30 minutes after a programming session for side effects, including: autonomic effects (e.g., facial flushing, facial muscle contractions, or increased heart rate), hypomania, increased disease symptoms, and sensations such as tingling, smell, or taste. For Obsessive-Compulsive Disorder, during treatment, patients should be monitored closely for increased depression, anxiety, suicidality, and worsening of obsessive-compulsive symptoms.

Patients should avoid activities that may put undue stress on the implanted components of the neurostimulation system. Activities that include sudden, excessive or repetitive bending, twisting, or stretching can cause component fracture or dislodgement that may result in loss of stimulation, intermittent stimulation, stimulation at the fracture site, and additional surgery to replace or reposition the component. Patients should avoid manipulating the implanted system components or burr hole site as this can result in component damage, lead dislodgement, skin erosion, or stimulation at the implant site. Patients should not dive below 10 meters (33 feet) of water or enter hyperbaric chambers above 2.0 atmospheres absolute (ATA) as this could damage the neurostimulation system, before diving or using a hyperbaric chamber, patients should discuss the effects of high pressure with their clinician.

Patients using a rechargeable neurostimulator for Parkinson's disease, essential tremor, dystonia, or epilepsy must not place the recharger over a medical device with which it is not compatible (eg, other neurostimulators, pacemaker, defibrillator, insulin pump). The recharger could accidentally change the operation of the medical device, which could result in a medical emergency. Patients should not use the recharger on an unhealed wound as the recharger system is not sterile and contact with the wound may cause an infection.

WARNING for Obsessive-Compulsive Disorder:

Electroconvulsive Therapy (ECT) – The safety of ECT in patients who have an implanted deep brain stimulation (DBS) system has not been established. Induced electrical currents may interfere with the intended stimulation or damage the neurostimulation system components resulting in loss of therapeutic effect, clinically significant undesirable stimulation effects, additional surgery for system explantation and replacement, or neurological injury.

PRECAUTIONS: Loss of coordination in activities such as swimming may occur. For Obsessive-Compulsive Disorder, the safety of somatic psychiatric therapies using equipment that generates electromagnetic interference (e.g., vagus nerve stimulation) has not been established. Patients using a rechargeable neurostimulator for Parkinson's disease, essential tremor, dystonia, or epilepsy should check for skin irritation or redness near the neurostimulator during or after recharging, and contact their physician if symptoms persist.

ADVERSE EVENTS: Adverse events related to the therapy, device, or procedure can include intracranial hemorrhage, cerebral infarction, CSF leak, pneumocephalus, seizures, surgical site complications (including pain, infection, dehiscence, erosion, seroma, and hematoma), meningitis, encephalitis, brain abscess, cerebral edema, aseptic cyst formation, device complications (including lead fracture and device migration) that may require revision or explant, extension fibrosis (tightening or bowstringing), new or exacerbation of neurological symptoms (including vision disorders, speech and swallowing disorders, motor coordination and balance disorders, sensory disturbances, cognitive impairment, and sleep disorders), psychiatric and behavioral disorders (including psychosis and abnormal thinking), cough, shocking or jolting sensation, ineffective therapy, and weight gain or loss.

For Parkinson's disease or essential tremor, safety and effectiveness has not been established for patients with neurological disease other than idiopathic Parkinson's disease or Essential Tremor, previous surgical ablation procedures, dementia, coagulopathies, or moderate to severe depression, patients who are pregnant, or patients under 18 years. For Essential Tremor, safety and effectiveness has not been established for bilateral stimulation or for patients over 80 years of age. For Dystonia, safety of this device for use in the treatment of dystonia with or without other accompanying conditions (e.g., previous surgical ablation procedure, dementia, coagulopathies, or moderate to severe depression, or for patient who are pregnant) has not been established. Age of implant is suggested to be that at which brain growth is approximately 90% complete or above. For Epilepsy, the safety and effectiveness of this therapy has not been established for patients without partial-onset seizures, patients who are pregnant or nursing, patients under the age of 18 years, patients with coagulopathies, and patients older than 65 years. For Obsessive-Compulsive Disorder, the safety and probable benefit of this therapy has not been established for patients with: Tourette's syndrome, OCD with a subclassification of hoarding, previous surgical ablation (e.g., capsulotomy), dementia, coagulopathies or who are on anticoagulant therapy, neurological disorders, and other serious medical illness including cardiovascular disease, renal or hepatic failure, and diabetes mellitus. In addition, the safety and probable benefit has not been established for these patients: those whose diagnosis of OCD is documented to be less than 5 years duration or whose YBOCS score is less than 30, who have not completed a minimum of 3 adequate trials of first and/or second line medications with augmentation, who have not attempted to complete an adequate trial of cognitive behavior therapy (CBT), who are pregnant, who are under the age of 18 years, and who do not have comorbid depression and anxiety. Physicians should carefully consider the potential risks of implanting the Reclaim DBS System in patients with comorbid psychiatric disorders (e.g., bipolar, body dysmorphic, psychotic) as the Reclaim DBS System may aggravate the symptoms.

***Humanitarian Device:** Authorized by Federal Law as an aid in the management of chronic, intractable (drug refractory) primary dystonia, including generalized and/or segmental dystonia, hemidystonia, and cervical dystonia (torticollis), in patients seven years of age or above. The effectiveness of the devices for treating these conditions has not been demonstrated. Authorized by Federal law for use as an adjunct to medications and as alternative to anterior capsulotomy for treatment of chronic, severe, treatment-resistant obsessive-compulsive disorder (OCD) in adult patients who have failed at least three selective serotonin reuptake inhibitors (SSRIs). The effectiveness of the devices for this use has not been demonstrated.

USA Rx only   Rev 09/22

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